NATURAL GAS VEHICLE (NGV) T-FITTING REPLACEMENT

AFFECTED VEHICLES: **CERTAIN 2002 MODEL YEAR CROWN VICTORIA NATURAL GAS VEHICLES (NGV)**

OVERVIEW

A ***new*** updated T-fitting must be installed on the No. 4 fuel tank. The valves on all of the fuel tanks must be locked down and any residual fuel pressure purged from the fuel lines prior to installing the ***new*** T-fitting. **The proper installation and tightening of the T-fitting is critical to ensure a leakproof repair. Therefore, you should read this document completely, including all notes, cautions and warnings before beginning work on the vehicle.**

WARNING!

THE FUEL SYSTEM OPERATES UNDER HIGH PRESSURE AND REMAINS PRESSURIZED AFTER ENGINE SHUTDOWN. THE FUEL SYSTEM PRESSURE MUST BE RELIEVED BEFORE ANY FUEL-RELATED COMPONENTS ARE SERVICED. FAILURE TO RELIEVE THE FUEL PRESSURE CAN RESULT IN PERSONAL INJURY OR PROPERTY DAMAGE.

WARNING!

DO NOT MODIFY THE SYSTEM CONFIGURATION OR COMPONENTS, OR INSTALL COMPONENTS OR PARTS NOT SPECIALLY DESIGNED FOR USE WITH NATURAL GAS. FORD MOTOR COMPANY HAS SPECIALLY DESIGNED THE MATERIALS, COMPONENTS AND SYSTEM CONFIGURATION FOR THE 4.6L CROWN VICTORIA NATURAL GAS VEHICLES, AND EACH PARTICULAR SYSTEM IS PRECISELY CALIBRATED FOR EFFICIENT OPERATION. THE USE OF DIFFERENT PARTS OR MATERIALS CAN PRODUCE AN UNTESTED CONFIGURATION THAT CAN RESULT IN FIRE OR PERSONAL INJURY, OR CAUSE ENGINE DAMAGE.

WARNING!

EYE AND EAR PROTECTION ARE REQUIRED TO BE WORN DURING VENTING OR REMOVAL AND INSTALLATION OF FUEL SYSTEM COMPONENTS. FAILURE TO FOLLOW THESE INSTRUCTIONS CAN RESULT IN PERSONAL INJURY.

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FUEL SYSTEM PRESSURE RELIEF

1. Install a memory saver, then disconnect the battery negative cable.
2. Remove the cover from the No. 3/4 tank vapor box. See Figure 1.
3. Close each manual lock-down valve on the No. 3 and No. 4 tanks. Tighten each lock-down valve to 8 Nm (71 lb-in). See Figure 1.
4. On vehicles equipped with a No. 5 tank, remove the access plug from the No. 5 tank vapor box cover. Using a long ball end hex bit, close the manual lock-down valve on the No. 5 tank. Tighten the lock-down valve to 8 Nm (71 lb-in). See Figure 1.

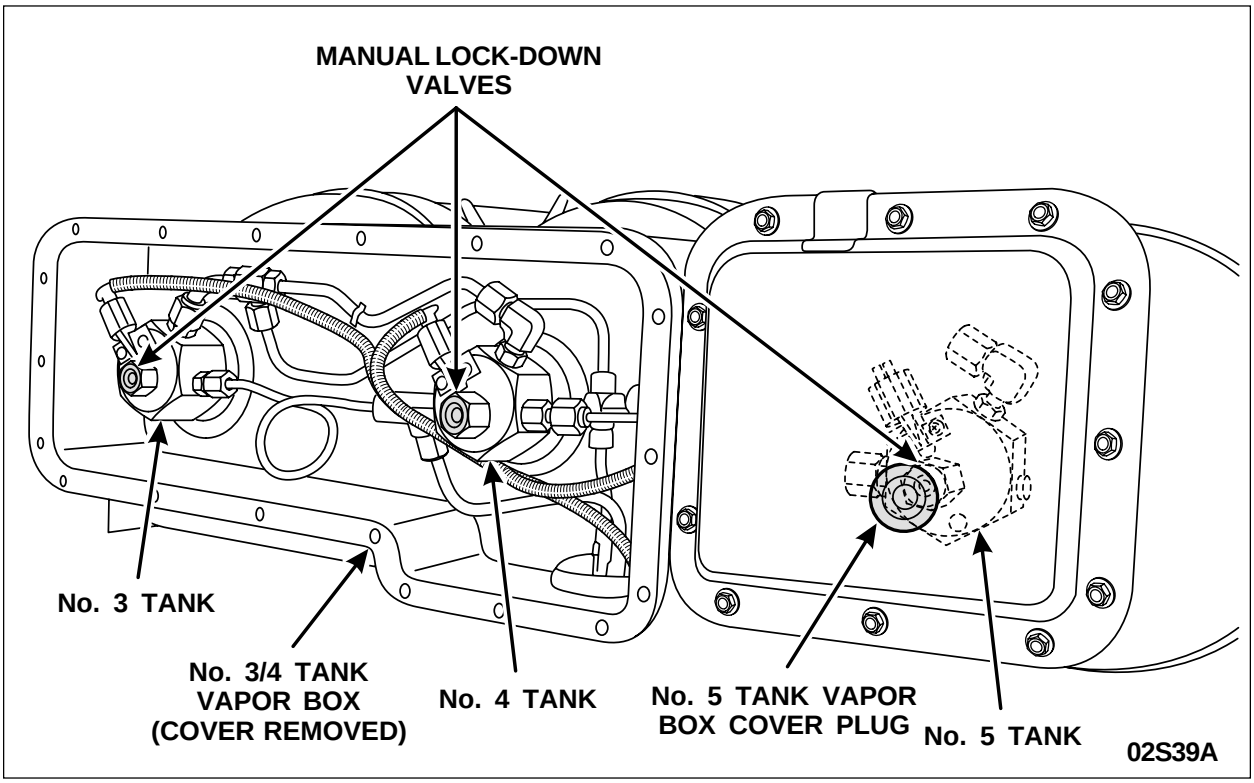


FIGURE 1

5. Raise the vehicle on a hoist.

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6. Remove the access plug from the No. 2 tank cover (located behind the left rear wheel). Close the manual lock-down valve on the No. 2 tank. Tighten the lock-down valve to 8 Nm (71 lb-in). See Figure 2.

7. Remove two (2) pushpins and position the rubber shield (located above left rear frame rail) for the No. 1 tank out of the way. Close the manual lock-down valve on the No. 1 tank. Tighten the lock-down valve to 8 Nm (71 lb-in). See Figure 2.

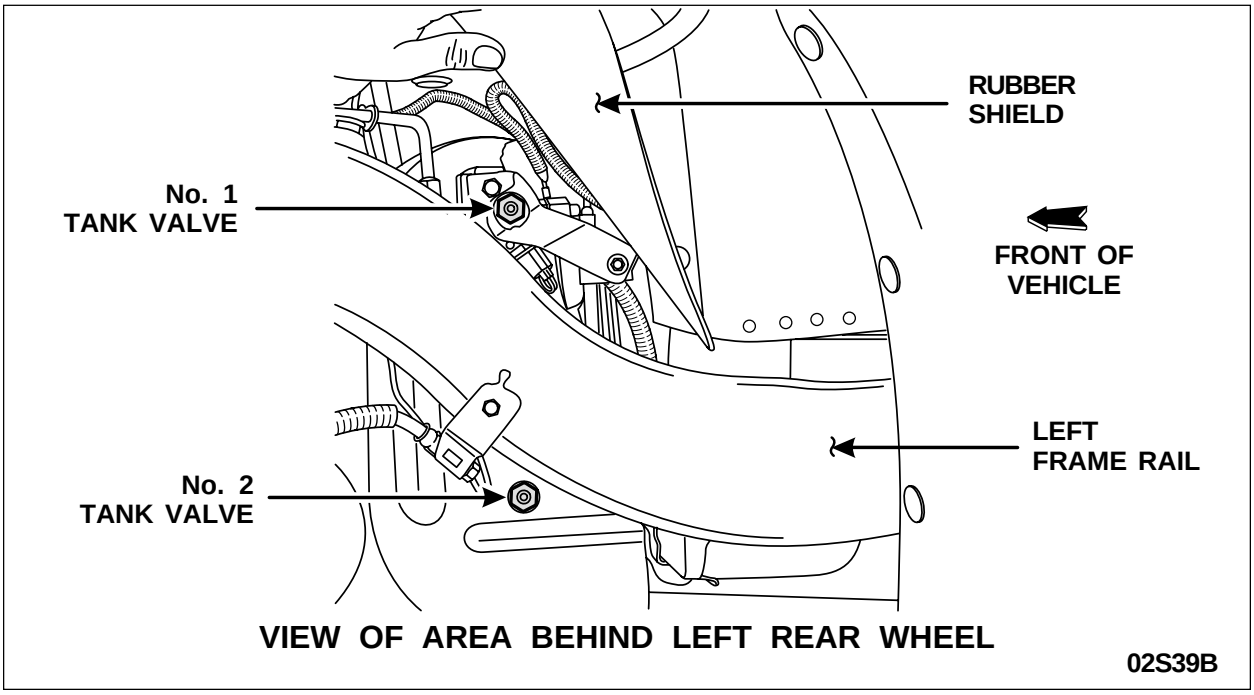


FIGURE 2

8. Lower the vehicle.

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9. Using the Rotunda Natural Gas (NGV) Pressure Test and Venting Kit (134-00114), relieve the fuel system pressure as follows.
   - Connect the grounding cable from the engine fuel supply manifold to the shop ground rod.
   - Connect the fuel rail pressure tester to the fuel supply manifold schrader valve.
   - Connect the venting hose assembly from the fuel rail pressure tester to the shop vent stack.
   - Slowly open the bleed valve on the fuel rail pressure tester and allow the contents of the fuel lines to vent to the atmosphere for one minute.
   - Close the bleed valve and remove the fuel rail pressure tester, venting hose assembly and grounding cable.
10. Slightly loosen the fuel line fitting on the No. 4 tank T-fitting to allow any residual pressure to escape from the system. See Figure 3.

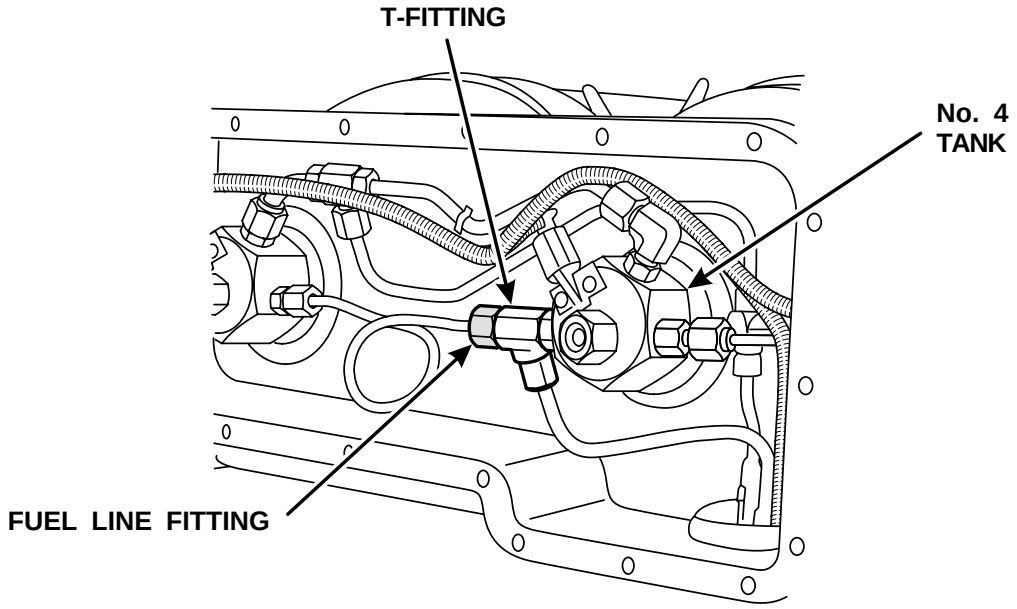


FIGURE 3

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T-FITTING REPLACEMENT

1. Disconnect the two (2) fuel lines (from lower tanks and No. 3 tank) from the T-fitting on the No. 4 tank. See Figure 4.
2. Loosen the locknut on the T-fitting, then remove and discard the T-fitting. See Figure 4.
3. **WARNING: DO NOT USE ANY SOLVENTS OR COMPRESSED AIR TO CLEAN ANY COMPONENTS OF THE NGV FUEL SYSTEM. SOLVENTS MAY DETERIORATE THE O-RINGS AND AFFECT THEIR SEALING CAPABILITIES. COMPRESSED AIR AND NATURAL GAS FORM A HIGHLY COMBUSTIBLE MIXTURE.**

   Clean the T-fitting mounting surface on the No. 4 tank with a clean dry cloth.

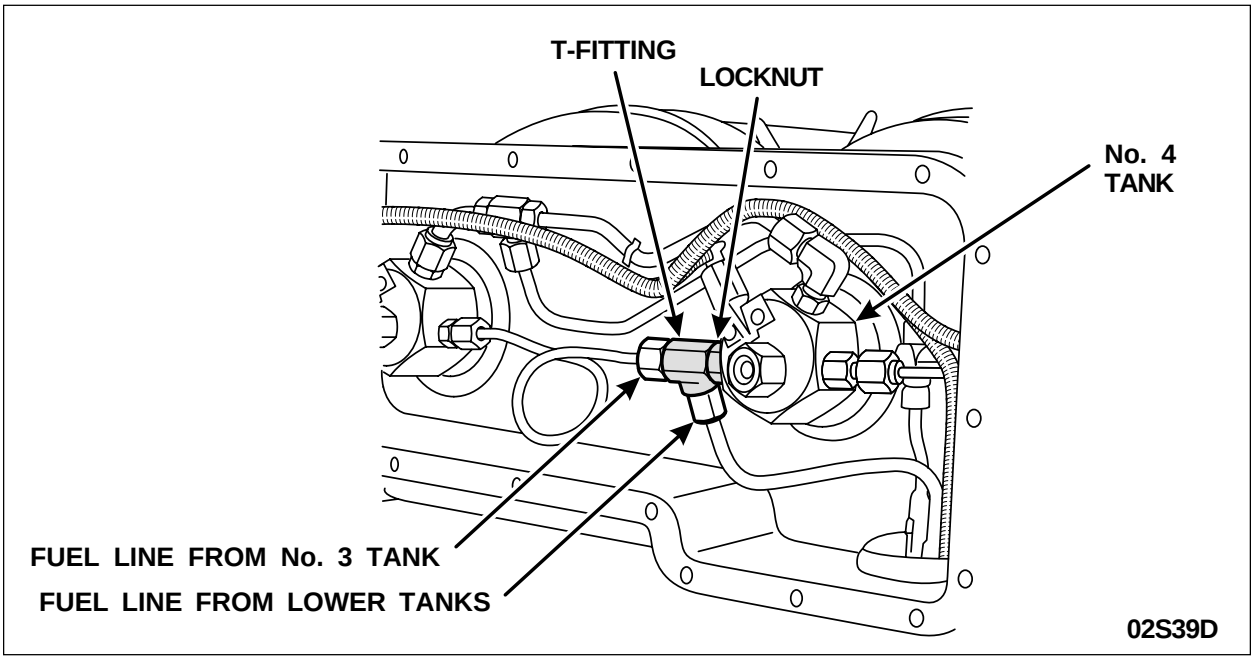


FIGURE 4

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4. **CAUTION: Apply the lubricant to the O-rings sparingly. Do NOT allow excess lubricant to touch the threads of the T-fitting. Any lubricant on the threads will adversely affect the tightening properties of the locknut and may damage the T-fitting during the tightening of the locknut.**

   Lubricate the O-rings on the ***new*** T-fitting with Motorcraft Super Premium SAE 5W-20 Motor Oil X0-5W20-QSP or equivalent meeting Ford specification WSS-M2C153-H. See Figure 5.

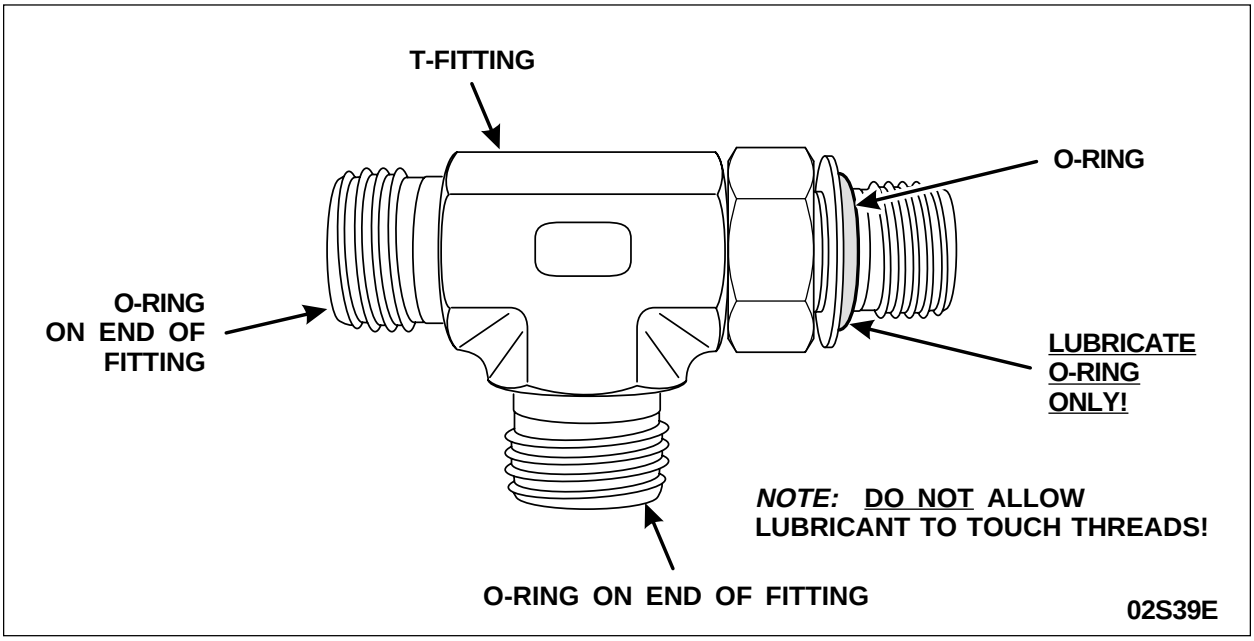


FIGURE 5

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5. Thread the ***new*** T-fitting into the No. 4 tank by hand until the O-ring makes contact. Continue tightening until the fuel line from the lower tanks is properly oriented with the ***new*** T-fitting. See Figure 6.

6. **CAUTION: The fuel lines must not contact the vapor box when they are in the installed position. If the fuel line contacts the vapor box, friction between the two components may result in wear of the fuel line and/or provide an ignition source for any vapors in the box.**

   Connect the fuel line from the lower tanks to the ***new*** T-fitting. Make sure the fuel line is not contacting the vapor box and adjust the position of the fuel line and/or T-fitting if necessary. Snug up the fuel line with a short open end wrench (do not final tighten at this time). See Figure 6.

7. Connect, but do not tighten, the fuel line from the No. 3 tank to the T-fitting. See Figure 6.

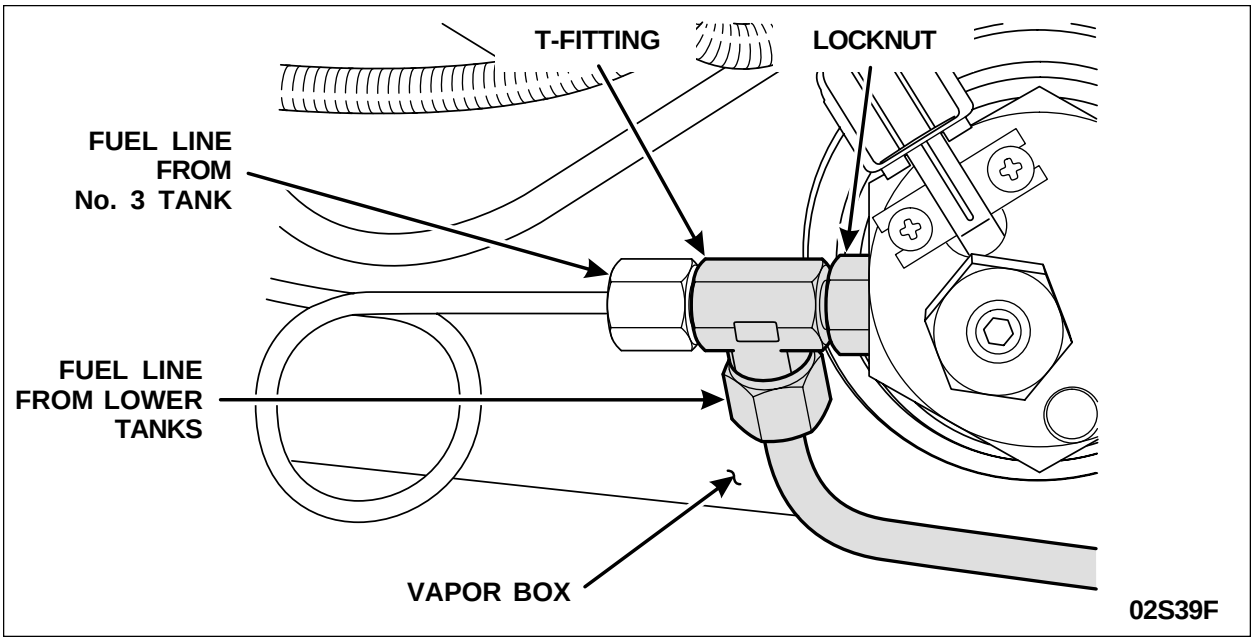


FIGURE 6

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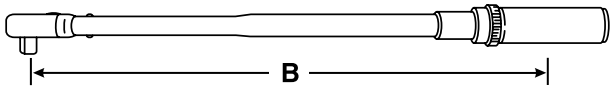
8. **CAUTION: The fuel lines must not contact the vapor box when they are in the installed position. If necessary, adjust the position of the T-fitting and/or the fuel line while tightening the components.**

   CAUTION: The use of a crow foot adapter on a torque wrench increases the length of the tool and changes the actual torque value. To achieve the proper torque on the fastener, you must use the formula supplied in this document to calculate the proper setting for the torque wrench/crow foot you are using.

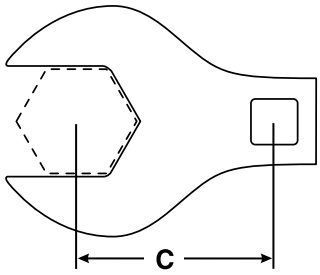
   Using the formula in Figure 7, calculate the proper torque setting, then tighten the T-fitting locknut and fuel line connections as follows.

   - While holding the ***new*** T-fitting in place, tighten the locknut with a crow foot adapter and a torque wrench to 43 Nm (32 lb-ft).
   - Tighten the fuel line from the lower tanks to the T-fitting to 43 Nm (32 lb-ft).
   - Tighten the fuel line from the No. 3 tank to the T-fitting to 43 Nm (32 lb-ft).

TORQUE WRENCH SETTING CALCULATOR



MEASURE FROM CENTER OF SQUARE DRIVE TO CENTER OF HANDGRIP WITH TORQUE WRENCH SET AT "0"



MEASURE FROM CENTERLINE OF SQUARE DRIVE TO CENTERLINE OF FASTENER

A = Desired torque of fastener
B = Torque wrench length measurement
C = Crowfoot Adapter length measurement
D = Required setting of torque wrench

$$\frac{A \times B}{B + C} = D$$

Example of typical tool application:

$$\frac{43 \text{ Nm} \times 14}{14 + 1.25} = \frac{602}{15.25} = 40 \text{ Nm}$$

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FIGURE 7

9. Raise the vehicle on a hoist.
10. Open the manual lock-down valve on the No. 1 tank. Tighten the lock-down valve in the open position to 8 Nm (71 lb-in). Reposition the rubber shield and install the two (2) pushpins.

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11. Open the manual lock-down valve on the No. 2 tank. Tighten the lock-down valve in the open position to 8 Nm (71 lb-in). Install the access plug into the No. 2 tank cover.
12. Lower the vehicle.
13. On vehicles equipped with a No. 5 tank, open the manual lock-down valve. Tighten the lock-down valve in the open position to 8 Nm (71 lb-in). Install the access plug into the No. 5 tank vapor box.
14. Open each manual lock-down valve on the No. 3 and No. 4 tanks. Tighten each lock-down valve in the open position to 8 Nm (71 lb-in).
15. Connect the battery negative cable, then remove the memory saver.
16. Turn the ignition to the RUN position to pressurize the fuel lines.
17. Leak check the repair by applying a soapy water solution to the T-fitting connections. Use of an inspection mirror will be necessary to view the connections. If any soap bubbles form, this indicates a leak at the connection. See Figure 8.
    - If a leak is found, turn off the ignition and call the Special Service Support Center (1-800-325-5621) for further instructions.
    - If no leaks are found, turn the ignition off and continue with this procedure.

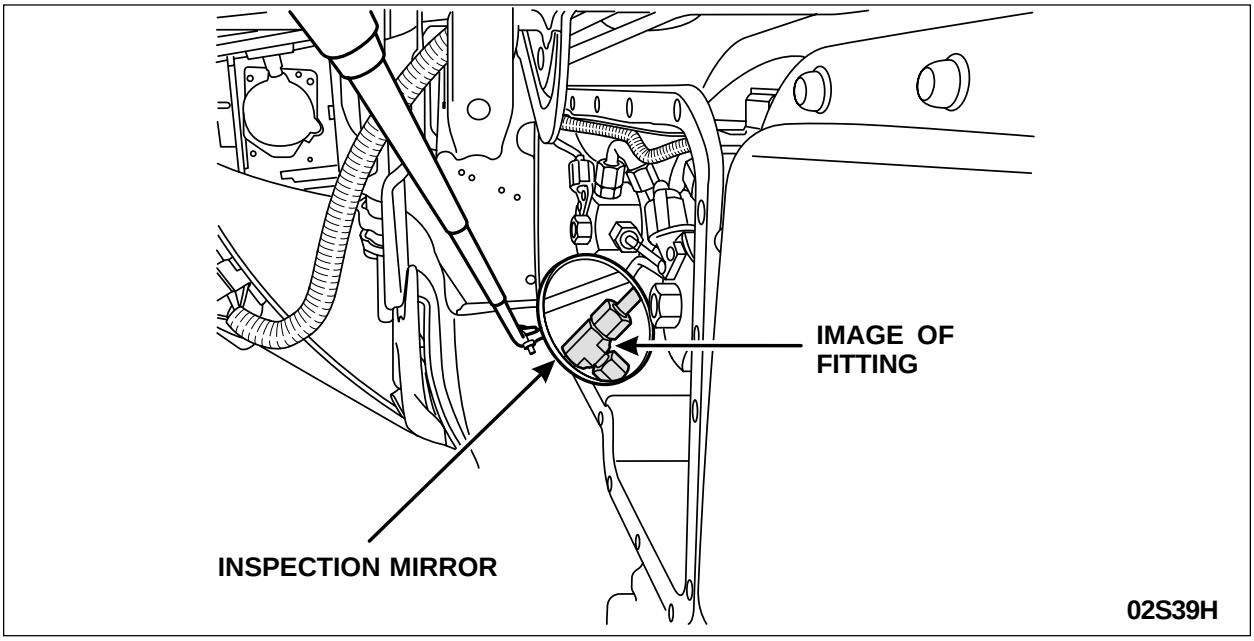


FIGURE 8

18. Install the cover on the No. 3/4 tank vapor box. Tighten the bolts to 9 Nm (80 lb-in).
19. Clear the diagnostic trouble codes.

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